ABSTRACT

Predicting NCAA March Madness Outcomes

Accurately predicting outcomes in the NCAA Division I basketball tournament presents a unique challenge due to its single elimination structure and the high variability inherent in short series competition. This study is motivated by the limitations of traditional bracket based approaches, which rely on binary selections and fail to capture uncertainty in game outcomes. This project investigates predictive modeling of tournament outcomes through statistical and machine learning approaches, with the goal of generating well calibrated win probabilities for every possible matchup.

Following a comprehensive review of existing literature, three modeling frameworks were constructed using R. These include a logistic regression model trained on Dean Oliver's Four Factors of Basketball Success, an XGBoost ensemble model incorporating tournament seeding, strength of schedule adjusted box score statistics, and a custom Elo rating system, and a Monte Carlo simulation framework that derives win probabilities from KenPom efficiency based point predictions. All models are evaluated using the Brier Score, which penalizes both incorrect and overconfident probability estimates. The findings indicate that ensemble methods incorporating dynamic rating systems provide a competitive edge in forecasting the high variance environment of single elimination tournament play.

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Predicting NCAA March Madness Outcomes

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We, the undersigned, certify that the project of the following student meets the required standards of scholarship, format, and style of the university and the student's graduate degree program for the awarding of the master's degree.

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INTRODUCTION

The National Collegiate Athletic Association (NCAA) serves as the premier developmental system for the National Basketball Association (NBA) and the Women's National Basketball Association (WNBA). With thousands of athletes across both divisions arriving from around the world, the influence of the NCAA is extensive. As the basketball world has evolved into a multibillion dollar industry [1], the value of a competitive edge has become substantial. Teams historically relied on live scouting and film study to gain these advantages. However, the last decade has seen a major shift toward statistical findings. Inspired largely by the movie *Moneyball*, teams now look to analytics to uncover value that the naked eye might miss.

The focal point of this competitive landscape is March Madness. At the conclusion of the regular season, the selection committee invites 68 teams to participate in this tournament. The field includes the champions from all 32 Division I conferences alongside the best remaining teams chosen at large. The committee seeds these participants into four distinct regions with rankings ranging from 1 to 16. The four teams judged to be the strongest are awarded the number 1 seeds while the next tier receives the number 2 seeds. Within each region, the teams compete in a single elimination format where matchups are initially determined by rank. For example, the top seed plays the sixteenth seed. The winner of each region advances to the Final Four to compete for the national championship. Crucially, these contests take place at neutral sites rather than on home courts. This structure creates a predictive challenge where volatility is the norm.

This project utilizes the Kaggle March Machine Learning Mania competition as its primary experimental framework. Kaggle provides a setting in which participants submit predicted win probabilities for every possible matchup, rather than making binary bracket selections. Although game outcomes are binary, this probabilistic formulation enables a more rigorous evaluation of model performance by assessing how closely predicted probabilities align with actual results. A model is therefore considered more accurate not because it assigns higher probabilities, but because its probability estimates are well calibrated, meaning that events predicted with higher probability occur more frequently in reality, while unlikely events are assigned appropriately low probabilities. This approach allows for a more nuanced assessment of predictive performance than traditional bracket-based methods. By using this structured setting, the research leverages standardized datasets, including team efficiency ratings and historical results, to develop a reproducible and statistically sound modeling pipeline.

This statistical consistency is possible because basketball is uniquely suited for machine learning applications. Both individual and collective performances are captured through a standardized set of metrics. Discrete variables such as points, steals, assists, and rebounds provide a consistent numerical foundation for analysis across different eras and levels of competition. This uniformity allows for the creation of structured datasets where team strength can be quantified and compared through objective performance indicators. By leveraging these patterns, machine learning models can identify underlying trends that govern game outcomes and team efficiency.

A basic assessment often relies on a record of wins and losses to infer the overall quality of a team. While these records serve as strong indicators, they frequently fail to capture critical nuance. Furthermore, the single elimination structure of the tournament ensures that upsets remain a constant possibility regardless of historical dominance. A defining example occurred in 2018 when the top ranked Virginia Cavaliers entered the postseason with 31 wins and only 3 losses. They fell to the lowest ranked UMBC Retrievers by 20 points. Such outcomes demonstrate why simple heuristics fail in this environment.

The NCAA Men's Basketball Tournament is defined by its structural unpredictability. The single-elimination format means that even the most statistically dominant teams face elimination from a single off night, and the compressed six-round schedule amplifies the role of momentum, fatigue, and matchup-specific advantages that aggregate metrics cannot fully capture. This variance is well-documented: research by Georgia Tech professor Joel Sokol, whose Logistic Regression/Markov Chain model has been benchmarked against nearly 100 ranking systems, finds that even the best predictive models achieve only approximately 75% accuracy, meaning roughly one in four tournament games results in an upset [29]. The consequences for bracket prediction are severe. DePaul mathematician Jeff Bergen calculates that the odds of a perfect 63-game bracket range from 1 in 9.2 quintillion under random selection to 1 in 128 billion for an informed basketball observer, and in the entire history of the tournament, no publicly verified bracket has ever survived past the Sweet Sixteen [28].

This inherent volatility drives a massive secondary market focused on sports betting. As legal wagering expands globally, March Madness has evolved into one of the largest gambling events in the world, with 3.1 billion wagered on tournament outcomes in 2025 [2]. To manage this risk, bookmakers generate odds that serve as the industry standard for predictive accuracy. These odds represent a market consensus that attempts to price in every available variable from player injuries to historical trends. However, research demonstrates that these lines are not purely probability-based: Simon (2024) finds that sportsbook forecasts overreact to incoming information and exhibit systematic inefficiencies inconsistent with weak-form market efficiency, suggesting that profit margins and bettor sentiment influence line-setting beyond pure probability [31]. Consequently, developing a model capable of outperforming these market-established baselines represents the gold standard for predictive success.

This paper aims to construct a predictive framework capable of meeting this standard. The primary objective is to develop and evaluate distinct statistical models that minimize the Brier Score, a proper scoring rule that measures the mean squared error between predicted probabilities and binary outcomes, making it particularly well suited for evaluating calibrated win probability estimates across tournament matchups [21]. By testing various algorithms against historical data, the research seeks to identify a strategy that generates a winning score on the Kaggle March Madness competition, which adopts the Brier Score as its evaluation metric. The final analysis benchmarks these models against competitor submissions that utilized multiple strategies including Elo ratings, standard odds conversion, and analytical rankings. Ultimately, the optimal model will be deployed in the upcoming 2026 March Madness tournament to compare its projections with live results.

Motivated by this challenge, substantial research has addressed the complexities of March Madness prediction. The majority of this work focuses on engineering feature sets for learning algorithms ranging from linear regression to more complex ensemble methods. While the Kaggle competition provides foundational game statistics, transforming these into effective predictors requires significant effort. The following subsection details the specific data types provided by the competition to establish this baseline. The final subsection we review existing literature with a primary focus on feature engineering. Finally, the paper details the implementation of distinct modeling techniques. This comparative analysis examines linear and logistic regression alongside additional statistical methods. These models are compared against one another and evaluated relative to the winning scores from prior competitions.

Dataset

The highly quantifiable nature of basketball ensures that statistical data is both abundant and accessible. While the Kaggle competition permits participants to incorporate any external datasets they acquire, the organizers themselves provide a comprehensive archive. This repository includes over 30 distinct tables of historical data spanning more than three decades. All tables are separated by gender, with the exception of the public rankings table, which is compiled exclusively for the men's division. The data provided by Kaggle is split into regular season and postseason files. For the purposes of this project, the tournament phase is referred to as the playoffs. This research focuses primarily on the detailed results tables for both the regular season and the playoffs while also utilizing the essential rankings and seeding tables. Although the broader archive extends further back, the detailed game statistics required for this analysis commence in the 2003 season. Each entry in these tables provides a comprehensive statistical breakdown for both the winning and losing teams in every recorded matchup. The schema utilizes the prefix `W` to denote statistics for the winning team and the prefix `L` for the losing team. Table 1 summarizes the specific variables recorded for each team. While raw box scores capture the foundational events of a game, they often

Table 1. Regular Season and Playoff Detailed Result Variables

Variable	**Description**
PTS	Points scored
FGM / FGA	Field goals made and attempted
FGM3 / FGA3	Three-point field goals made and attempted
FTM / FTA	Free throws made and attempted
OR / DR	Offensive and defensive rebounds secured
Ast	Assists
TO	Turnovers
Blk	Blocks
PF	Personal fouls

fail to account for nuances such as pace of play and strength of schedule. To address this limitation, this research supplements the Kaggle datasets with advanced metrics from analyst Ken Pomeroy (KenPom), who is widely accepted as the industry standard for college basketball analysis. Rather than relying on raw counting statistics, Pomeroy normalizes performance by calculating efficiency on a per possession basis. These ratings are updated daily throughout the regular season and tournament. For the purposes of this study, a snapshot of the rankings was acquired from the day prior to the start of the 2025 tournament. Table 2 describes the primary metrics utilized from this source [10]. A possession is defined as the period in which a team controls

Table 2. KenPom Metrics

Metric	Description
NetRtg	Differential between Offensive and Defensive Rating
ORtg	Points scored per 100 possessions, adjusted for opponent
DRtg	Points allowed per 100 possessions, adjusted for opponent
AdjT	Number of possessions per 40 minutes

the ball, ending with a shot attempt, turnover, or score. Because Pomeroy's ratings are exclusively available for the men's division, this research incorporates an additional dataset from Bart Torvik for the women's division. Torvik cites Pomeroy as a significant influence, and the ratings are constructed using a comparable methodology. However, distinct algorithmic differences exist between the two systems; for further detail regarding these distinctions, the reader is referred to [3]. Table 3 describes the metrics utilized from this source. To complement the efficiency metrics provided by Pomeroy

Table 3. Bart Torvik Metrics (Women's Division)

Metric	Description
ADJOE	Adjusted offensive efficiency
ADJDE	Adjusted defensive efficiency
AdjT	Adjusted tempo

and Torvik, this research also incorporates the comprehensive ranking systems aggregated by Kenneth Massey. While the previously discussed datasets focus on predictive efficiency, the Massey Ordinals provide a consolidated view of team hierarchy by aggregating rankings from over 40 distinct mathematical systems, pollsters, and algorithms. Rankings are presented as ordinal integers where the top team is assigned the value of 1. Incorporating this data allows the models to leverage a "wisdom-of-the-crowd" approach by drawing on the consensus of the broader analytical community. One significant challenge in aggregating these distinct sources is the inconsistency in team nomenclature. For instance, a university might be listed as St. Mary's" in one dataset and Saint Mary's CA" in another. To resolve this, all external data was mapped to the unique `TeamID` integer provided by the Kaggle competition files, ensuring that advanced efficiency metrics are accurately merged with the box score data for every matchup.

LITERATURE REVIEW

Past Strategies

The pursuit of a perfect bracket has generated a substantial body of research in predictive modeling over the last decade. While specific algorithms vary, the most successful strategies in the Kaggle Machine Learning Mania competition consistently rely on one of two distinct methodological frameworks.

The primary objective for this study is to minimize the Brier Score. We define the Brier Score (BS) as follows:

$$BS = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2 \tag{1}$$

where N is the number of events, f_t is the predicted probability, and o_t is the actual outcome. For this research, a 0 is considered a loss, and a 1 is a win. A Brier Score of 0 would reflect perfect accuracy, while a Brier Score of 1 reflects perfect inaccuracy. Table 4 illustrates how the score behaves across a range of prediction scenarios.

Table 4. Brier Score Examples

Predicted	**Outcome**	**Scenario**	**Brier Score**
0.50	Win or Loss	baseline	0.2500
1.00	Win	Perfectly confident, correct	0.0000
1.00	Loss	Perfectly confident, incorrect	1.0000
0.70	Win	Moderately confident, correct	0.0900
0.70	Loss	Moderately confident, incorrect	0.4900
0.90	Win	Highly confident, correct	0.0100
0.90	Loss	Highly confident, incorrect	0.8100

The table highlights an important property of the Brier Score: overconfident incorrect predictions are penalized more severely than moderate ones. For instance, predicting a 90% win probability for a team that ultimately loses yields a score of 0.81, far worse than the 0.25 baseline. This incentivizes well-calibrated probability estimates rather than extreme predictions.

The first framework relies on fundamental analysis where the objective is to construct a ground-truth probability independent of public sentiment. A common implementation of this approach involves utilizing domain-specific heuristics such as the elo rating system. Originally designed for chess, elo provides a dynamic measure of team strength that updates after every game. It offers a robust baseline that accounts for strength of schedule more effectively than raw win-loss records [13]. Many strategies also incorporate expert features. These include the Nate Silver, FiveThirtyEight, and KenPom ratings which act as high-signal inputs that aggregate player health, travel distance, and roster depth into a single numeric value. By treating these expert outputs as features rather than final predictions, modelers can leverage human domain expertise while correcting for known expert biases.

The second dominant framework relies on the Efficient Market Hypothesis [9]. This approach posits that global betting markets, specifically closing lines from major bookmakers, offer a more accurate aggregate predictor of game outcomes than individual statistical models. The efficacy of this strategy is well-documented in recent Kaggle competitions where five of the top eight performing models utilized market-derived features as their primary feature. These strategies do not attempt to beat the market. Instead, they focus on mathematically disentangling the true win probability from the bookmaker's overround and inherent biases. To achieve this separation, recent state-of-the-art implementations utilize specific conversion algorithms to de-noise betting data. The standard tool in this domain is the `goto_conversion` algorithm. Competitors employ these functions to address the favorite-longshot bias, which is the tendency for markets to overvalue underdogs and undervalue favorites. Raw Vegas odds refer to the moneyline prices posted by sportsbooks, which embed both the bookmaker's profit margin and systematic biases from public betting behavior. By applying conversion corrections to these raw figures, modelers can transform them into calibrated win probabilities that minimize Brier Score more effectively than fundamental analysis alone. This approach effectively leverages the crowd wisdom of the market while stripping away the pricing inefficiencies introduced by public betting behavior. For more information on how this strategy works, we refer the reader to [12].

The applicability of these methodological frameworks, however, is heavily constrained by the structural discontinuity introduced by the 2021 Name, Image, and Likeness (NIL) legislation. A comparative analysis of historical team winning percentages in the post-NIL landscape reveals a significant deviation from historical norms. This shift is characterized by a rapid consolidation of talent where the number of programs maintaining elite win percentages ($> 75\%$) has nearly tripled compared to the preceding two decades [6]. This phenomenon implies that feature weights derived from pre-2021 eras are likely subject to significant concept drift. Consequently, this literature review restricts its focus to winning strategies from recent Kaggle competitions (2022-present), as methodologies developed before the NIL era may not reflect the approaches best suited to the heightened stratification of team strength driven by current transfer portal dynamics.

Modeling Techniques

Recent tournament prediction methodologies employ several established modeling and computational techniques. This section provides formal definitions for the five core approaches used in this study: Logistic Regression, Linear Regression, Monte Carlo Simulation, Random Forest, and Neural Networks.

Logistic Regression is a binary classification model that estimates the probability of an outcome by applying the logistic (also known as sigmoid) function to a linear combination of input features. The model is defined as:

$$P(Y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2+\ldots+\beta_n x_n)}} \tag{2}$$

where Y is the binary outcome, $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ represents the feature vector, and $\boldsymbol{\beta} = (\beta_0, \beta_1 \ \ldots, \beta_n)$ are the model coefficients estimated through the maximum likelihood estimation. This function ensures the predicted probabilities are bounded between $[0, 1]$. Logistic regression assumes a linear relationship between the input features and the log-odds of the outcome. This assumption can be restrictive when the true relationships are nonlinear. To address this limitation, common approaches are feature engineering with polynomial or interaction terms (e.g, $x^2, x_i \cdot x_j$) to capture nonlinear patterns.

Linear Regression is a supervised learning method that models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data. Linear Regression is defined as:

$$\hat{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k \tag{3}$$

where $\hat{Y}$ is the predicted continuous outcome, $X = (x_1, x_2, \ldots, x_k)$ represents the vector of features, and $\beta = (\beta_0, \beta_1, \ldots, \beta_k)$ are the model coefficients estimated through ordinary least squares (OLS) minimization. For more information on OLS we refer the reader to [14]. Unlike logistic regression, linear regression produces unbounded predictions and does not inherently output probabilities. In sports prediction contexts, linear regression is commonly used to estimate point totals, scoring margins, or other continuous performance metrics that can subsequently be transformed into win probabilities through additional techniques, such as applying a logistic transformation to the predicted margin or feeding the output into a Monte Carlo simulation to estimate the probability that a given margin corresponds to a win.

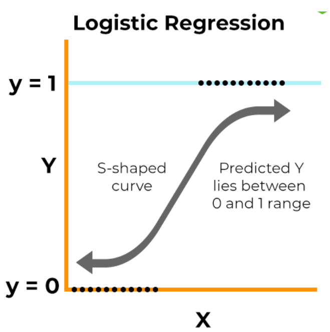

Figure 1. An Example of a Logistic Regression Equation [15].

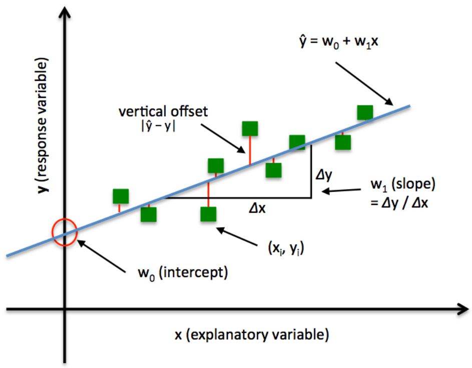

Figure 2. An Example of a Linear Regression Equation [17].

Monte Carlo Simulation is a computational technique that uses repeated random sampling to estimate probabilities for systems with inherent uncertainty. The method generates N independent samples $x_1, x_2, \ldots, x_N$ from a probability distribution $f(x)$, then estimates probabilities by computation the proportion of samples satisfying a certain condition. The simulation is defined as:

$$\hat{P} = \frac{1}{N} \sum_{i=1}^{N} 1(x_i \in A) \tag{4}$$

where $1(\cdot)$ is the indicator function that equals 1 when the condition is satisfied and 0 otherwise. A represents the event of interest, and $\hat{P}$ is the Monte Carlo estimate of the true probability $P(A)$. By the Law of Large Numbers [18], $\hat{P} \to P(A)$ as $N \to \infty$. In sports prediction contexts, Monte Carlo methods are frequently employed to simulate game outcomes by sampling from estimated score distributions, enabling the estimation of win probabilities through empirical frequency counts across simulated matchups.

XGBoost (Extreme Gradient Boosting), considered a "black-box" approach, is an ensemble learning method that constructs a sequence of decision trees where each successive tree is trained to correct the residual errors of the previous trees. Unlike Random Forest, which builds trees independently, XGBoost employs gradient boosting to iteratively minimize a loss function through additive modeling [5]. The model prediction is defined as the sum of predictions from all trees:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i). \tag{5}$$

Here, $f_k(x_i)$ is the prediction of the k^{th} tree for the i^{ith} data point, $\hat{y}_i$ is the final predicted value for the i^{ith} data point, and K is the number of trees. Each tree is trained to minimize the objective function

$$\text{obj}(\theta) = \sum_{i}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{6}$$

where l is the loss function which simply computes the difference between the true and predicted, and Ω is the regularization term, discouraging overly complex trees. The optimization process then proceeds via gradient descent on the residuals. For more information regarding gradient descent we refer the reader to [4]. XGBoost is widely adopted in competitive machine learning for its performance on structured data and ability to capture complex interactions and relationships between features.

PROBABILITY PREDICTION MODELS

This section details the development and evaluation of models designed to predict a continuous win probability $P \in (0, 1)$ for any given matchup. While the tournament results are ultimately binary, the Brier Score evaluation metric heavily penalizes overconfidence in incorrect predictions and rewards well-calibrated probabilities. Consequently, this research moves beyond simple classification to focus on models that quantify the uncertainty inherent in the March Madness environment. By establishing a framework where a value $P > 0.5$ denotes a predicted victory for Team 1, a baseline is created for performance that accounts for the high variance of single-elimination play. This probabilistic approach serves as the engine of the project and bridges the gap between theoretical literature and the empirical application of machine learning.

To expand further on these concepts, the methodology begins with a rigorous data processing phase where historical box scores are transformed into a symmetric training set. This ensures that the models learn to evaluate the relative strength between two competitors rather than relying on arbitrary formatting artifacts. This section details the complete machine learning pipeline implemented for the 2025 tournament prediction. We first examine the data cleaning procedures required to tailor the feature sets to the specific statistical assumptions of each algorithm. Following this, we explore the variable selection process used to identify high signal metrics, including tournament seeding, opponent-adjusted box score statistics, strength of schedule, and Elo ratings derived from historical game results. Finally, we implement and compare three distinct modeling architectures—logistic regression, XGBoost, and a Monte Carlo simulated linear regression model—to determine which approach best minimizes the Brier Score in a high-variance, single-elimination environment.

Logistic Regression

Logistic regression serves as the natural starting point for this analysis due to its direct alignment with the structure of the prediction problem. The outcome variable is binary, a team either wins or loses, and logistic regression models the log-odds of this outcome as a linear combination of input features, producing a well-calibrated probability as output without requiring additional transformation. Its interpretability is also a key advantage at this stage: the estimated coefficients provide direct insight into which features drive the predicted probability, establishing a transparent and theoretically grounded baseline against which the more complex models can be compared.

Variable Selection

To accurately model basketball outcomes, one must first deconstruct the sport into its atomic statistical events. A standard game is composed of a finite sequence of possessions. Within these possessions, specific discrete actions determine the flow of the contest. The primary events recorded in the raw dataset include Field Goals, Free Throws, Rebounds, Turnovers, and Personal Fouls. A Field Goal (FGM) represents the primary method of scoring and is worth either two or three points. A Free Throw (FTM) is an unopposed attempt worth one point awarded after a foul. A Rebound is recorded when a player retrieves the ball after a missed shot; this is further categorized into Offensive Rebounds (OR), which grant the shooting team a second opportunity, and Defensive Rebounds (DR), which terminate the opponent's possession. Finally, a Turnover (TO) is a loss of possession due to a violation or steal, resulting in zero points. Analyzing these events using raw totals is statistically flawed due to the confounding variable of pace. A team that plays at a high tempo may record high totals for points and rebounds simply because they generate more possessions, not because they are efficient. To resolve this, this model utilizes the "Four Factors of Basketball Success." Developed by Dean Oliver [20], which isolates the four components that correlate most strongly with winning: shooting efficiency, turnover avoidance, rebounding dominance, and free throw frequency.

Central to these calculations [16] is the estimation of total possessions (Poss), which must be derived from box score data using the following formula:

$$Poss = (FGA - OR) + TO + (0.44 \cdot FTA) \tag{7}$$

The first factor is Effective Field Goal Percentage (eFG%), which adjusts for the added value of three-point shots:

$$eFG\% = \frac{FGM + 0.5 \cdot FGM3}{FGA} \tag{8}$$

The second factor is Turnover Percentage (TOV%), which normalizes turnovers against total possessions:

$$TOV\% = \frac{TO}{Poss} \tag{9}$$

The third factor is Offensive Rebound Percentage (ORB%), which measures rebounding dominance relative to available opportunities:

$$ORB\% = \frac{OR}{OR + DR_{opp}} \tag{10}$$

Finally, the model incorporates Win Percentage (Win%) as a fifth feature to capture intangible qualities such as coaching and clutch performance, calculated as the mean of binary game results over the course of the season. All features are transformed into differentials between the two competing teams prior to modeling, centering the analysis on the relative mismatch between opponents rather than their absolute statistics.

Data Preparation

To construct a robust predictive pipeline, the data architecture must establish a strict boundary between feature generation and model training. The independent variables, specifically the Four Factors and Win Percentage, are calculated exclusively utilizing historical regular-season detailed results. Conversely, the logistic regression algorithm is trained entirely on historical NCAA tournament outcomes. By isolating regular-season efficiency metrics as the predictors and tournament game results as the target variable, the framework ensures that the model evaluates how a team's sustained performance over a thirty-game span translates to the high-variance, single-elimination environment of postseason play. This distinction is critical, as regular-season data provides a sufficiently large sample size to stabilize efficiency metrics, while tournament data provides the specific contextual outcomes the model ultimately aims to predict.

Furthermore, to prevent data leakage and simulate the exact informational constraints of a real-world forecaster, the logistic regression model is trained using a strictly defined rolling time window. To evaluate historical accuracy and construct valid out-of-sample predictions, the framework is systematically back tested across consecutive tournament years. For example, to generate probability forecasts for the 2025 NCAA Tournament, the model is trained exclusively on the tournament matchups from the preceding 2023 and 2024 seasons. The decision to restrict the training data to a brief temporal window is grounded in the structural realities of collegiate athletics. NCAA basketball is characterized by high roster turnover, as student-athletes possess limited eligibility and typically serve as primary contributors for only three to four seasons. Consequently, a university program undergoes a complete personnel and strategic transformation within a short cycle. Training the algorithm on data exceeding this window would introduce noise from past iterations of a team that share no roster overlap with the current squad. During this localized training phase, the algorithm maps the recent regular-season metrics to their respective tournament results, learning the optimal coefficient weights without accessing any data from the target prediction year. This rolling training window shifts iteratively for historical benchmark predictions, ensuring that all evaluations remain mathematically rigorous and entirely devoid of look-ahead bias.

The primary obstacle in processing the historical game records is the fundamental structure of the Kaggle dataset. The raw regular-season and tournament data files utilize a deterministic outcome schema. Within this structure, every game observation is explicitly bifurcated into winning and losing statistics. Metrics belonging to the victorious program are assigned column headers prefixed with a 'W', while the defeated opponent's metrics are prefixed with an 'L'.

If a classification algorithm is trained directly on this format, it inevitably suffers from severe target leakage. The logistic regression algorithm would trivially learn to associate the column position with the outcome, mathematically recognizing that the entity occupying the 'W' feature space possesses a 100 percent win probability. This renders the model entirely incapable of out-of-sample prediction, as the ultimate objective is to forecast matchups between two neutral entities before the outcome is known. The following output illustrates this problematic data architecture, demonstrating how the game result is inherently encoded into the feature presentation itself.

```
head(mens_results %>%
       select(Season, WTeamID, WScore,
              LTeamID, LScore, WFGM,
              LFGM, WTO, LTO), 4)

##    Season WTeamID WScore LTeamID LScore  WFGM  LFGM   WTO   LTO
##     <int>   <int>  <int>   <int>  <int> <int> <int> <int> <int>
## 1:   2003    1104     68    1328     62    27    22    23    18
## 2:   2003    1272     70    1393     63    26    24    13    12
## 3:   2003    1266     73    1437     61    24    22    10    12
## 4:   2003    1296     56    1457     50    18    18    12    19
```

Table 5. Snapshot of raw Kaggle regular-season data demonstrating the 'W' and 'L' column structure.

To resolve the target leakage inherent in the raw dataset, the first phase of data processing transforms the game-level observations into season-level team averages. This transformation is executed via the `calculate_four_factors` function. Initially, the function estimates the total possessions for both the winning and losing teams using the standard formula. Subsequently, it computes the Four Factors (Effective Field Goal Percentage, Turnover Percentage, Offensive Rebound Percentage, and Free Throw Rate) for each team in every game.

Rather than maintaining the winner and loser statistics in a single row, the function separates the victors and the defeated into two distinct dataframes. A binary `Win` variable is appended, assigning a value of 1 to the winning dataframe and 0 to the losing dataframe. These two structures are then vertically concatenated using a row-bind operation. Finally, the data is grouped by `Season` and `TeamID` to calculate the mean of all numeric variables. This yields a single, stable efficiency profile and an aggregate regular season win percentage for every program in a given year.

Model Implementation

Following the systematic cleaning and neutralization of the dataset, the analysis proceeds to the implementation of the predictive framework. The primary objective is to construct a logistic regression pipeline capable of generating calibrated win probabilities for every possible matchup in the 2026 NCAA Tournament. This method is chosen specifically for its ability to model the probability of a binary outcome, in this case whether the lower-ID team (Team 1) wins or loses, based on a linear combination of statistical differentials. The models utilize a logit link function to map the relationship between the independent variables and the probability of a victory. The formal specification for each model iteration is defined by the following equation:

$$\text{logit}(P) = \ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1(\text{eFG\_diff}) + \beta_2(\text{TOV\_diff}) + \beta_3(\text{ORB\_diff}) + \beta_4(\text{FTR\_diff}) + \beta_5(\text{Win\_diff}) \tag{11}$$

Where P represents the probability of a Team 1 victory. The coefficients (β) represent the change in log-odds of winning for every one unit increase in the respective statistical differential.

The model is estimated using maximum likelihood estimation, which identifies the coefficient vector β that maximizes the probability of observing the recorded tournament outcomes given the input differentials. Because the response variable is binary, the log-likelihood function takes the form:

$$\sum_{i=1}^{n} [y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i))] \tag{12}$$

where $y_i \in \{0, 1\}$ is the observed outcome for the i-th matchup and $p(x_i)$ is the model's predicted probability [22] of a Team 1 victory. Maximizing this function is equivalent to minimizing the binary cross-entropy loss. The resulting coefficients reflect both the direction and magnitude of each differential's contribution to the probability of winning. A positive coefficient on `eFG_diff`, for instance, indicates that a larger shooting efficiency advantage is associated with a higher probability of victory, which is consistent with what basketball theory would predict.

For the 2026 deployment, separate models are trained for the men's and women's divisions. The choice to train independent models for each division rather than a single combined model with a gender indicator reflects a fundamental assumption that the Four Factors carry structurally different predictive weight in the men's and women's game. Differences in pace, physicality, and three-point volume between the two divisions suggest that the relationship between shooting efficiency and winning probability may not be uniform across genders. Training separate models allows each set of coefficients to reflect the unique dynamics of its respective division rather than forcing a shared parameter estimate that could mask meaningful differences.

Nonetheless, Each model is trained exclusively on tournament outcomes from the 2024 and 2025 seasons, with the corresponding regular-season Four Factors serving as the independent variables. This two-year window was selected to reflect the current competitive landscape of college basketball while respecting the structural roster turnover discussed previously. The Kaggle competition provides a submission file containing every possible 2026 tournament matchup, numbering in the thousands of pairings across both divisions. The Four Factors for each team are computed from their 2025 regular-season results and differenced for every candidate pairing. The fitted models then generate a probability $\hat{P} \in (0, 1)$ for each of these matchups via the inverse logit transformation, producing the final submission.

An important practical consideration in this implementation is the relatively small sample size introduced by the two-year training window. A single NCAA tournament produces only 63 games for the men's bracket and

63 for the women's, yielding a combined training pool of roughly 250 observations per model iteration. While this is sufficient for a low-dimensional model such as logistic regression, it places meaningful constraints on the complexity of the architecture that can be reliably estimated. Overfitting becomes a nontrivial risk if additional interaction terms or polynomial features are introduced without sufficient regularization. For this reason, the model is deliberately kept simple, relying on five theoretically motivated differentials rather than a broader set of engineered features. This simplicity is consistent with the principle of Occam's Razor in model selection [23], where a simpler model that generalizes well out-of-sample is preferred over a complex one that fits the training data too closely.

Prior to submitting predictions for the 2026 tournament, the pipeline is validated through a series of backtests spanning the 2023, 2024, and 2025 competitions. These historical evaluations serve a specific purpose: to confirm that the rolling two-year training window generalizes to unseen tournament data before any live predictions are committed. For the 2023 backtest, the model is trained on tournament outcomes from 2021 and 2022. For the 2024 backtest, the training window shifts forward to include 2022 and 2023. Finally, for the 2025 backtest, the model is trained on 2023 and 2024 tournament data. In each case, the Four Factors are derived exclusively from the regular-season data preceding the target year, and no tournament results from the prediction year are ever introduced into the training process.

The resulting Brier Scores from these backtests are compared against the winning submission scores published by Kaggle for the corresponding competition years. This comparison provides a meaningful external benchmark, situating the logistic regression framework within the broader landscape of competitor strategies. Table 6 summarizes the Brier Scores produced by the model across all three backtest years alongside the winning scores for their respective competitions.

Table 6. Logistic Regression Backtest Results vs. Kaggle Benchmarks

Tournament Year	Model Brier Score	Winning Brier Score
2023	0.21653	0.17371
2024	0.21064	0.05313
2025	0.18401	0.10411

It is important to note that the 2024 winning Brier Score of 0.05313 is not directly comparable to the scores recorded in 2023 and 2025. In that competition year, Kaggle introduced a significant structural change to the submission format. Rather than submitting a single set of game-by-game win probabilities, competitors were permitted to submit a portfolio of up to one hundred thousand distinct bracket entries. The final score was then computed by averaging the Brier Score across rounds rather than across individual games, placing substantially greater weight on later-round matchups where fewer games are played. This round-weighted averaging mechanism naturally compresses scores toward lower values relative to the standard per-game formulation. Consequently, the 0.05313 figure reflects a fundamentally different evaluation environment and should be interpreted as a benchmark specific to the 2024 format rather than as evidence of superior predictive accuracy over other competition years.

Setting aside the 2024 anomaly, the remaining backtests reveal a consistent trend of improvement. The model's Brier Score declined from

0.21653 in 2023 to 0.18401 in 2025, reflecting incremental gains in calibration as the pipeline was refined. Crucially, the 2025 backtest represents the most direct analog to the 2026 submission, as it uses the identical training structure and feature set that will be deployed. While the model trails the top competitors in each year, it remains well above the naive 0.25 baseline and demonstrates stable generalization across multiple tournament cycles. These results provided sufficient confidence in the pipeline to justify its deployment for the 2026 Kaggle submission.

Model Analysis

With the models estimated on the 2023 and 2024 tournament data, the resulting coefficients can be examined to evaluate which of the Four Factors carry meaningful predictive weight in each division. Because separate models were trained for the men's and women's tournaments, the coefficient structures can be analyzed independently and then compared to assess whether the Four Factors operate differently across divisions. Tables 7 and 8 present the estimated coefficients, p-values, and significance levels for the men's and women's models respectively.

Table 7. Logistic Regression Coefficient Estimates (Men's 2025 Model)

Variable	Estimate	p-value	Sig.
Intercept	0.4352	0.0348	*
eFG Diff	7.7952	0.2244	
TOV Pct Diff	-20.5106	0.0901	.
ORB Pct Diff	12.3634	0.0027	**
FTR Diff	-1.4307	0.7527	
Win Diff	1.3253	0.4365	

Table 8. Logistic Regression Coefficient Estimates (Women's 2025 Model)

Variable	Estimate	p-value	Sig.
Intercept	-0.0888	0.6657	
eFG Diff	17.4458	0.0016	**
TOV Pct Diff	-19.8212	0.0237	*
ORB Pct Diff	12.4308	0.0004	***
FTR Diff	3.9742	0.4420	
Win Diff	-1.8468	0.4172	

In the men's model, the only variable to reach conventional statistical significance is Offensive Rebound Percentage, with an estimated coefficient of 12.3634 and a p-value of 0.003. The positive sign is theoretically consistent with Dean Oliver's framework [20], as a team that dominates the offensive glass generates additional possessions while simultaneously denying the opponent easy transition opportunities. Turnover Percentage approaches significance at the 0.10 level with an estimate of -20.5106, and the negative direction is again theoretically correct, as a team that surrenders possessions more frequently than its opponent is expected to win less often. Effective Field Goal Percentage, Free Throw Rate, and Win Percentage all fail to reach significance in the men's model, suggesting that within the narrow two-year training window, shooting efficiency differentials and win records carry limited additional explanatory power once rebounding and turnover control are accounted for. The intercept of 0.4352 is statistically significant at the 0.05 level, implying a slight structural tendency for the lower-ID team to win in the men's tournament even after controlling for all efficiency differentials. This is likely an artifact of the team ID assignment convention rather than a genuine basketball phenomenon.

The women's model tells a different story. Three variables reach statistical significance: Offensive Rebound Percentage (12.4308, $p < 0.001$), Effective Field Goal Percentage (17.4458, $p = 0.002$), and Turnover Percentage (-19.8212, $p = 0.024$). The consistency of Offensive Rebound Percentage across both models is the single most robust finding of this analysis, as the near identical magnitude of the estimates (12.36 in the men's model and 12.43 in the women's model) across two independently trained models strongly suggests that second chance scoring dominance is a universal determinant of tournament success regardless of division. The women's intercept is effectively zero and insignificant, indicating no structural bias toward either team ordering in that division.

The most substantive structural difference between the two models is the role of Effective Field Goal Percentage. In the women's model, it is the second most significant predictor with a coefficient nearly twice the magnitude of the men's estimate. In the men's model, the same variable is statistically indistinguishable from zero. This divergence suggests that shooting efficiency differentials are a more decisive factor in women's tournament outcomes, where a meaningful gap in $eFG\%$ between two opponents translates more reliably into a victory. In the men's tournament, the compressed talent distribution and faster pace of play may reduce the marginal impact of a given efficiency advantage, as superior athleticism and shot creation ability can compensate for efficiency deficits in ways that are not captured by the Four Factors alone. This finding provides direct empirical support for the decision to train the two models independently rather than pooling them under a shared parameter structure. Had a combined model been used, the women's strong shooting efficiency signal would have been diluted by the men's null result, and the men's rebound signal would have been the only surviving predictor across both divisions.

Free Throw Rate fails to reach significance in either model, which is consistent with the nature of the statistic itself. Unlike shooting efficiency or turnover rate, free throw volume tends to be relatively stable across teams at the tournament level, as elite programs generally reach the line at comparable rates. When the differential between two opponents is consistently narrow, the variable carries little discriminatory power and the model appropriately assigns it a negligible coefficient. Win Percentage similarly contributes no significant explanatory power in either model once the efficiency differentials are controlled for. This suggests that the Four Factors effectively subsume the information contained in raw win-loss records; a team's season-long efficiency profile captures the underlying quality of the program more precisely than its aggregate record, which can be inflated or deflated by schedule strength and margin of victory.

Results

The performance of the 2025 logistic regression models is evaluated along two dimensions: classification accuracy via the confusion matrices and probability calibration via the Brier Score. Figure 3 presents the confusion matrices for the men's, women's, and combined models.

The women's model achieves the strongest classification performance at 77.6% accuracy, followed by the combined model at 73.9% and the men's model at 70.1%. This ordering is consistent with the coefficient analysis in the

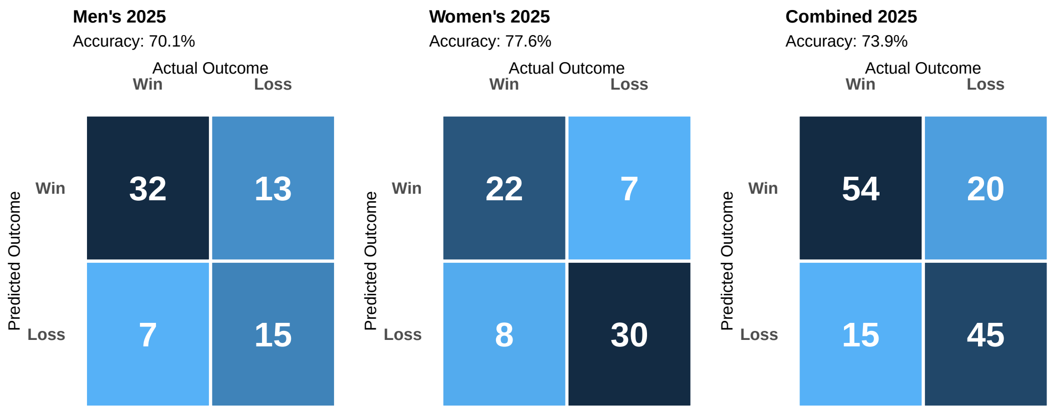

Figure 3. Confusion Matrices for the 2025 Logistic Regression Models

prior section. The women's model benefits from three statistically significant predictors, Offensive Rebound Percentage, Effective Field Goal Percentage, and Turnover Percentage, providing a richer signal for binary classification. The men's model, where only Offensive Rebound Percentage reaches conventional significance, has a narrower basis for discrimination and consequently misclassifies a greater proportion of games. All three accuracy figures comfortably exceed the 50% baseline that would be expected from random prediction, indicating that the Four Factors carry genuine discriminatory power in both divisions.

However, classification accuracy alone is an incomplete measure of model quality in this context. The Brier Score evaluation framework penalizes not just incorrect predictions but overconfident ones, meaning a model that correctly identifies the winner but assigns a probability of 0.95 rather than 0.65 is still penalized heavily when upsets occur. The combined 2025 Brier Score of 0.18401 reflects this tension. While the model demonstrates solid directional accuracy, the gap between its Brier Score and the competition winning score of 0.10411 suggests that the predicted probabilities are not as well calibrated as those produced by the top competitors. This is an expected limitation of a parsimonious four factor framework. Models that incorporate additional signals such as KenPom efficiency ratings, seed differentials, or market-derived probabilities are better positioned to assign more conservative probabilities to matchups involving significant uncertainty, which is precisely where the Brier Score penalizes most heavily.

Despite this gap, the 2025 results validate the pipeline as a functional and reproducible baseline. The model generalizes consistently across both divisions, produces accuracy well above chance, and maintains a Brier Score substantially below the naive 0.25 baseline. These characteristics provide a foundation from which the more sophisticated modeling techniques discussed in the following sections aim to improve.

Boosting

While logistic regression provides an interpretable and theoretically grounded baseline, its core assumption of a linear relationship between the input differentials and the log-odds of winning imposes meaningful constraints on its predictive ceiling. In reality, the relationship between team efficiency metrics and tournament success may be nonlinear and subject to complex interactions. For instance, a team with a dominant offensive rebounding advantage may only convert that advantage into wins when paired with a sufficient shooting efficiency threshold, a conditional relationship that a linear model cannot capture without explicit feature engineering. To address this limitation, the analysis extends the modeling framework to include XGBoost.

XGBoost is a natural candidate for this problem for several reasons beyond its flexibility. Unlike parametric models, it does not require assumptions about the distributional form of the predictors or their relationship to the outcome. It is also robust to the scale differences between features, meaning the large coefficient magnitudes observed for Turnover Percentage and Offensive Rebound Percentage in the logistic regression output do not require normalization before fitting. Furthermore, XGBoost includes built-in regularization parameters that directly penalize model complexity, which is particularly valuable given the small training sample introduced by the two-year rolling window. The following subsections detail the variable selection, data cleaning, and implementation decisions made for the boosting framework, mirroring the structure of the logistic regression analysis to facilitate a direct comparison between the two approaches.

Variable Selection

A key motivation for the breadth of this feature set is that XGBoost's internal regularization framework absorbs the cost of including additional variables in a way that ordinary regression cannot. In logistic regression, adding correlated or weakly predictive features inflates standard errors, destabilizes coefficient estimates, and risks overfitting, which is why the Four Factors model was deliberately restricted to five predictors. XGBoost does not share this vulnerability. At each boosting round the algorithm selects only the features and split points that produce the greatest reduction in the loss function, effectively ignoring features that contribute no signal. Features that are redundant or noisy are simply passed over during tree construction rather than distorting the model as they would in a linear framework. This property is reinforced by the `colsample_bytree` hyperparameter, which randomly samples a subset of features at each tree, further reducing the influence of any single uninformative predictor. As a result, the XGBoost feature set can be expanded aggressively without the same penalty that would apply in a parametric setting, and the cost of including a feature that turns out to be irrelevant is negligible compared to the cost of omitting one that carries genuine signal.

The XGBoost model is trained on all NCAA tournament games between 2003 and 2024, combining both the men's and women's divisions into a single dataset. This differs from the logistic regression approach, where the divisions are modeled separately. The decision to combine divisions here is deliberate: XGBoost is a high-capacity model with a substantially larger feature set, and merging both divisions approximately doubles the number of available training observations. Given that the total number of tournament games in any single division across 22 seasons is modest relative to the complexity of the feature space, this pooling strategy reduces the risk of overfitting and allows the model to learn more stable feature relationships. The underlying assumption is that the predictive structure of tournament outcomes, specifically the relationships between efficiency differentials, seeding, and win probability, is sufficiently consistent across divisions to justify training on a combined sample. To ensure the model can still account for any systemic, structural differences between the two tournaments, a binary `men_women` indicator variable is retained in the feature set. This allows the algorithm to learn universal interactions across the pooled data while applying bracket-specific adjustments when necessary. The choice of 2003 as the starting point reflects the availability of detailed box score statistics in the Kaggle data, as earlier seasons contain only compact results without the possession-level statistics needed to compute the strength of schedule adjusted features. Season identifiers, team identifiers, and games played counts are excluded from the feature matrix entirely, as they carry no generalizable predictive signal and would cause the model to memorize season-specific patterns rather than learn relationships that transfer to future tournaments.

The feature set for the XGBoost model is organized into three tiers of increasing complexity: seeding features, simple season statistics, and strength of schedule adjusted box score profiles. Together these tiers provide the model with information about committee evaluation, raw performance, and opponent-adjusted efficiency. The first tier consists of tournament seeding. Each team's seed number is included individually as `T1_seed` and `T2_seed`, along with their difference `Seed_diff`, defined as the opponent's seed minus the team's seed such that a positive value indicates a favorable matchup. Including all three seed features rather than the differential alone allows the model to learn nonlinear effects of absolute seeding. A seed differential of one means something very different when it separates a 1 seed from a 2 seed than when it separates an 8 seed from a 9 seed. The lower seeded teams in the tournament are generally far more dominant over their opponents than the differential alone would suggest, and a model that only sees the difference cannot distinguish between these cases. Figure 4 confirms this through two complementary views. The left panel shows that average point differential declines monotonically as seed number increases across both divisions, while the right panel shows that as the seed differential between two teams widens, the better seeded team wins by larger margins on average, validating seed differential as a strong predictive signal in its own right.

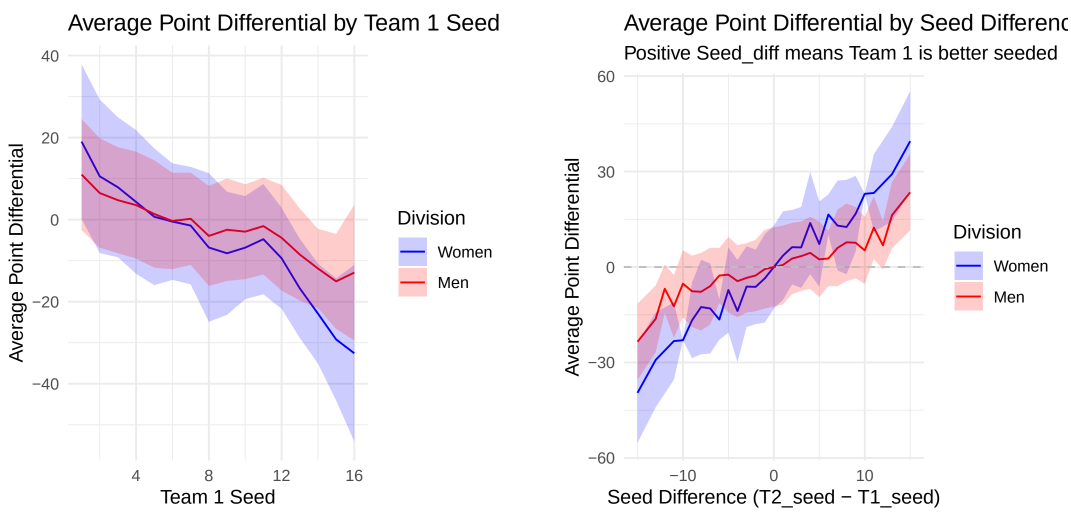

Figure 4. Left: Average point differential by Team 1 seed. Right: Average point differential by seed differential.

The second tier consists of simple season statistics computed from regular season game logs. For each team these include season win percentage, average points scored, average points allowed, and average point differential. Pairwise differentials between Team 1 and Team 2 are also computed for each of these statistics, providing the model with both absolute team quality signals and direct matchup comparison signals. These features capture overall team strength but do not account for the quality of opposition faced. A team that averaged 85 points per game against a weak conference schedule is not necessarily more dangerous than a team that averaged 75 points against a gauntlet of top 25 opponents, and the raw averages alone cannot make that distinction. This limitation motivates the third tier.

The third tier introduces strength of schedule adjusted box score profiles. For each of the fourteen raw box score statistics available in the Kaggle data, including field goals made and attempted, three point attempts, free throw attempts, offensive and defensive rebounds, assists, turnovers, steals, blocks, and personal fouls, seasonal averages are computed from two perspectives. The first perspective captures a team's own offensive and defensive production, prefixed as `Off_` and `Def_` respectively. The second perspective captures the quality of opposition faced by averaging the seasonal profiles of every opponent a team played during the regular season, producing a set of `SOS_` prefixed features. A team whose offensive statistics were produced against consistently strong defensive opponents is credited more heavily than a team whose statistics were inflated by a weak schedule. All box score statistics are adjusted for overtime prior to averaging using the formula $\texttt{adjOT} = \frac{40+5\times\texttt{NumOT}}{40}$, which scales each game's statistics back to a standard 40 minute equivalent. This ensures that high scoring overtime games do not artificially inflate a team's seasonal averages.

The final component of the feature set is a custom Elo rating system developed to capture dynamic team strength across the full regular season. Unlike the static season averages described in the previous tiers, Elo ratings are updated after every game and reflect the cumulative result of a team's performance relative to its opponents throughout the season. This means a team that starts slowly but peaks heading into the tournament will have a higher Elo rating than its season averages alone would suggest, and conversely a team that fades late in the season will have a deflated rating relative to its win percentage.

The Elo system used here is adapted from the rating system originally developed for chess by Arpad Elo [13], with the implementation structure inspired by [7], originally written in Python and reimplemented in R for this analysis. Each team begins the first season with an initial rating of 1500. After each regular season game the winner gains rating points and the loser loses an equal number, preserving the total rating mass in the system. The magnitude of the exchange depends on how surprising the result was relative to the pre-game expectations implied by the two teams' current ratings. The expected win probability for Team 1 is computed as:

$$E_1 = \frac{1}{1 + 10^{(R_2 - R_1)/400}} \tag{13}$$

where R_1 and R_2 are the current ratings of Team 1 and Team 2 respectively. The denominator of 400 is the width parameter that controls how sensitive the expected probability is to rating differences. After the game resolves, ratings are updated as:

$$R_1 = R_1 + k(1 - E_1)$$
$$R_2 = R_2 + k(0 - E_2)$$

where $k = 64$ is the K-factor controlling the maximum possible rating change per game and $E_2 = 1 - E_1$. When a heavy favorite wins, E_1 is close to 1 and the rating change is small since the result was expected. When a heavy underdog wins, E_1 is close to 0 and the rating change is large. This property makes Elo particularly well suited for tournament prediction because late season upsets and dominant runs cause large rating movements that static averages cannot capture. A key design decision in this implementation is the between-season carry over. Rather than resetting every team to 1500 at the start of each new season, the end of season rating is partially carried forward according to:

$$R_{\texttt{new}} = 0.5 \times R_{\texttt{final}} + 750. \tag{14}$$

This means 50% of a team's historical rating is preserved into the next season while the other 50% regresses toward the mean. The motivation is that program quality is not purely year to year. A program like Duke or

Connecticut that has been consistently elite for two decades should not start each season at the same rating as a mid-major program making its first tournament appearance. The carry over allows the Elo system to accumulate this historical signal while the regression toward the mean prevents ratings from drifting arbitrarily far from the initial value over many seasons. Figure 5 illustrates the end-of-season Elo ratings for the top 20 men's and women's programs in 2025, with the left panel displaying the men's rankings and the right panel displaying the women's rankings, reflecting both current season performance and accumulated historical quality. Each team's final end-of-season Elo rating is included in the feature set as `T1_Elo` and `T2_Elo`, along with their difference `Elo_diff`. Including all three allows the model to learn both absolute and relative rating effects simultaneously.

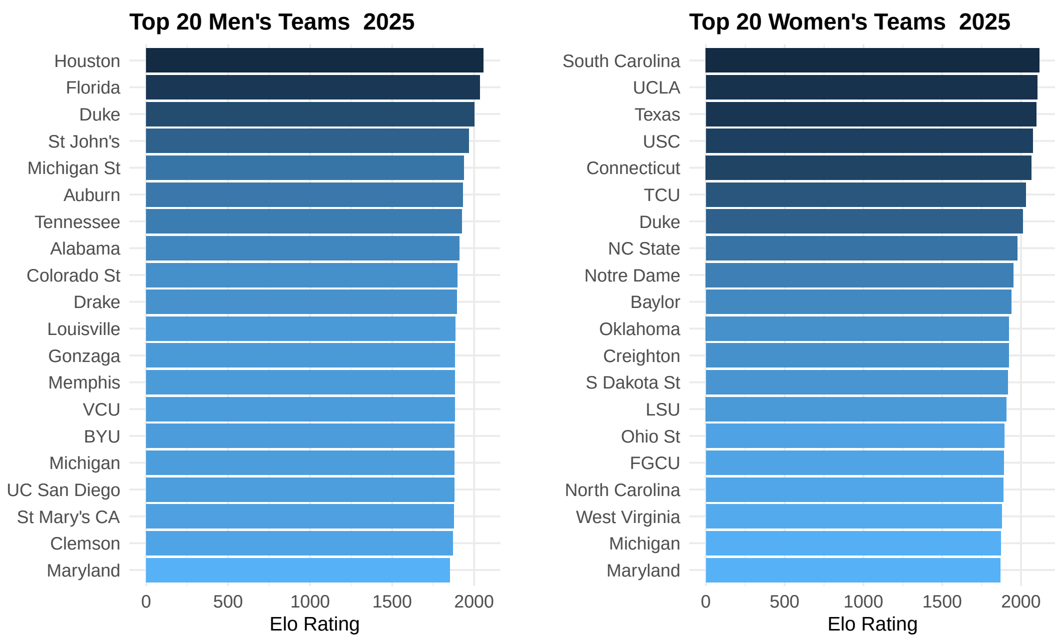

Figure 5. Left: Top 20 men's programs by end of season Elo rating in 2025. Right: Top 20 women's programs by end of season Elo rating in 2025.

Data Preparation

A critical design decision in the data construction is the symmetric duplication of each tournament game. For every game in the dataset, two rows are created: one where the winning team is assigned to the T1 position and one where the losing team is assigned to the T1 position. The binary outcome variable is set to 1 when T1 wins and 0 when T1 loses. This duplication serves three purposes. First, it allows the model to learn from both the winner's and loser's perspective in every matchup, meaning the model sees what a strong team's feature profile looks like when it wins and what a weak team's feature profile looks like when it loses. Second, it ensures that the model does not develop a positional bias toward the T1 slot; since every game appears from both orientations, the model cannot exploit any systematic advantage of being labeled T1. Finally, it guarantees an exactly balanced target distribution (50% wins, 50% losses), preventing the model from developing a bias toward a majority class. This duplication doubles the effective size of the training data, which is particularly valuable given that the total number of tournament games across all seasons is modest relative to the complexity of the feature set. Prior to feature computation, all box score statistics are scaled to a standard 40 minute equivalent using the overtime adjustment defined in the Variable Selection section, ensuring that high scoring overtime games do not artificially inflate seasonal averages.

Once all features are computed, the training matrix is filtered to retain only rows where every feature is present using `complete.cases`. This step is included as a safeguard against missing values that could arise if a team appeared in the tournament without sufficient regular season history to compute stable averages, or if the Elo system had not yet accumulated enough games to generate a meaningful rating. In practice no rows were removed, indicating that every tournament team in the dataset had sufficient regular season history to produce a complete feature profile across all seasons from 2003 to 2024. This is partly a consequence of the 2003 cutoff, which ensures that every season in the training data has a full year of regular season results available before the tournament begins. The Elo ratings included as features are computed from the full regular season history spanning 2003 to 2025, with between-season carry-over applied at each transition, meaning the 2025 end-of-season ratings reflect accumulated performance across all seasons in the training window rather than the current season alone. The final curated training matrix consisted of $4,552$ rows and 139 features.

The 2025 test set is constructed to exactly mirror the training data pipeline. Every feature engineering step, merge, and column ordering applied during training is repeated identically for the test set. This is a critical requirement for tree-based models since XGBoost maps each input column to a specific feature index at prediction time, and any mismatch between the training and test feature matrices will produce incorrect predictions. Once the test matrix is verified to match the training structure, the model generates a win probability for each 2025 tournament matchup, which is then evaluated against the actual outcomes.

Model Implementation

With the training matrix assembled and the feature set finalized, the focus shifts to training the XGBoost model and selecting the configuration that produces the best out of sample performance. Two key decisions must be made before training can begin: the number of boosting rounds, which determines how many sequential trees are added to the ensemble, and the hyperparameter values that control the complexity and regularization of each individual tree. Both decisions have a meaningful impact on model performance and neither can be determined from the training data alone without risking overfitting. To address this, the model uses five fold cross validation to guide both the round selection and the hyperparameter tuning process.

Cross validation is a resampling technique used to evaluate model performance on data that was not used during training. The core motivation is that evaluating a model on the same data it was trained on produces an overly optimistic estimate of performance, since the model has already seen and fit to that data. A naive solution is to hold out a fixed portion of the data as a test set, but with a small dataset this sacrifices training observations that could meaningfully improve the model. Cross validation addresses this trade off by rotating which portion of the data is held out across multiple iterations, allowing every observation to serve as both a training example and a validation example at different points. In $k-$fold cross validation the dataset is partitioned into k equally sized subsets called folds. The model is trained k times, each time using $k-1$ folds as the training set and the last fold as the validation set [27]. The performance metric is recorded for each fold and the final cross validated score is the average across all k folds. This produces a more reliable estimate of out of sample performance than a single train test split because the result is averaged over multiple different held out subsets rather than depending on one particular partition of the data. For more comprehensive information on cross validation methodology, we refer the reader to [26].

In this analysis five fold cross validation is applied to the full tournament training dataset to determine the optimal number of boosting rounds. At each fold the model is trained on 80% of the tournament games and evaluated on the held out 20%, with the average Brier Score across all five folds used as the selection criterion. Early stopping is applied with a patience of 50 rounds, meaning training halts automatically if the cross validated Brier Score does not improve for 50 consecutive rounds, preventing the model from overfitting by adding trees that contribute no meaningful reduction in error. Cross validation identified 415 as the optimal number of boosting rounds, at which point the model achieved its lowest average cross validated Brier Score across the five folds. Five folds was chosen as a balance between bias and variance in the performance estimate. Using fewer folds, such as three, means each validation set is larger but the training set is smaller, which can underestimate the model's true performance on a full dataset. Using more folds, such as ten, produces a training set that more closely resembles the full dataset but increases computational cost and can produce high variance estimates when the dataset is small.

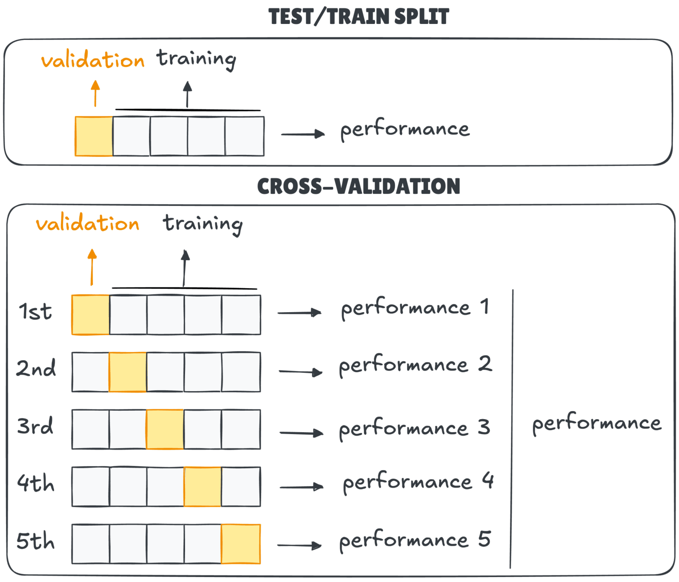

Figure 6. Example of Cross Validation and Regular Test/Train Split [24]

Model Analysis

The model was trained utilizing a logistic regression objective function for binary classification, evaluating performance directly via a custom Brier Score metric. To maximize computational efficiency across the expanded feature space, a histogram-based tree method was employed, and a static random seed was set to guarantee reproducibility. In addition to the number of boosting rounds, five hyperparameters were tuned to control the complexity and regularization of the model. Each parameter was tuned independently by fixing all others at their current best value and testing candidate values one at a time, selecting the value that produced the lowest cross validated Brier Score before moving to the next parameter.

The learning rate `eta` controls how much each new tree contributes to the ensemble. A lower value of `eta` requires more boosting rounds to converge but produces a smoother and more regularized model. A value of `eta = 0.01` was selected after testing values of 0.02 and 0.005, with 0.01 producing the lowest cross validated Brier Score. The `max_depth` parameter controls the maximum depth of each individual tree, limiting how many sequential splits a tree can make and therefore how complex the interactions it can capture. A value of `max_depth = 5` was selected after testing values of 4 and 6. The `subsample` parameter controls the fraction of training rows randomly sampled when building each tree, introducing stochasticity that reduces overfitting. A value of `subsample = 0.8` was selected, meaning each tree is built on a random 80% of the training data. Similarly, `colsample_bytree = 0.7` means each tree randomly samples 70% of the available features, further reducing the influence of any single predictor and encouraging the model to find robust signals across the full feature set. Finally, `min_child_weight = 4` sets the minimum number of training observations required in a leaf node before a split is made, preventing the model from creating highly specific splits that apply to only a handful of games. Together these five parameters define a model that is expressive enough to capture nonlinear tournament dynamics while remaining sufficiently regularized to generalize to unseen data.

The feature importance output from the final model ranks each predictor by its average gain across all splits, where gain measures the reduction in the Brier Score attributable to each feature. As shown in Figure 7, `Seed_diff` is the most influential predictor by a wide margin, accounting for 32.2% of the total gain. This is consistent with the exploratory analysis, which showed a strong relationship between seed differential and point differential, and confirms that tournament seeding remains the single strongest signal for predicting matchup outcomes. `Elo_diff` ranks second with 11.5% of the total gain, capturing cumulative season long performance in a way that seeds alone cannot. Together these two features account for roughly 44% of the model's predictive power. `avg_point_diff_diff` contributes an additional 5.5%, reflecting the margin by which each team outperformed its opponents during the regular season. Beyond these three features, the remaining predictors each contribute less than 3% individually, with strength of schedule variables such as `T1_SOS_PointDiff`, `T2_SOS_win`, and `T2_SOS_Off_Blk` making up the next tier, followed by a long tail of box score statistics. The concentration of predictive power in seeding and Elo suggests that broad measures of team quality drive tournament outcomes more than any specific box score metric.

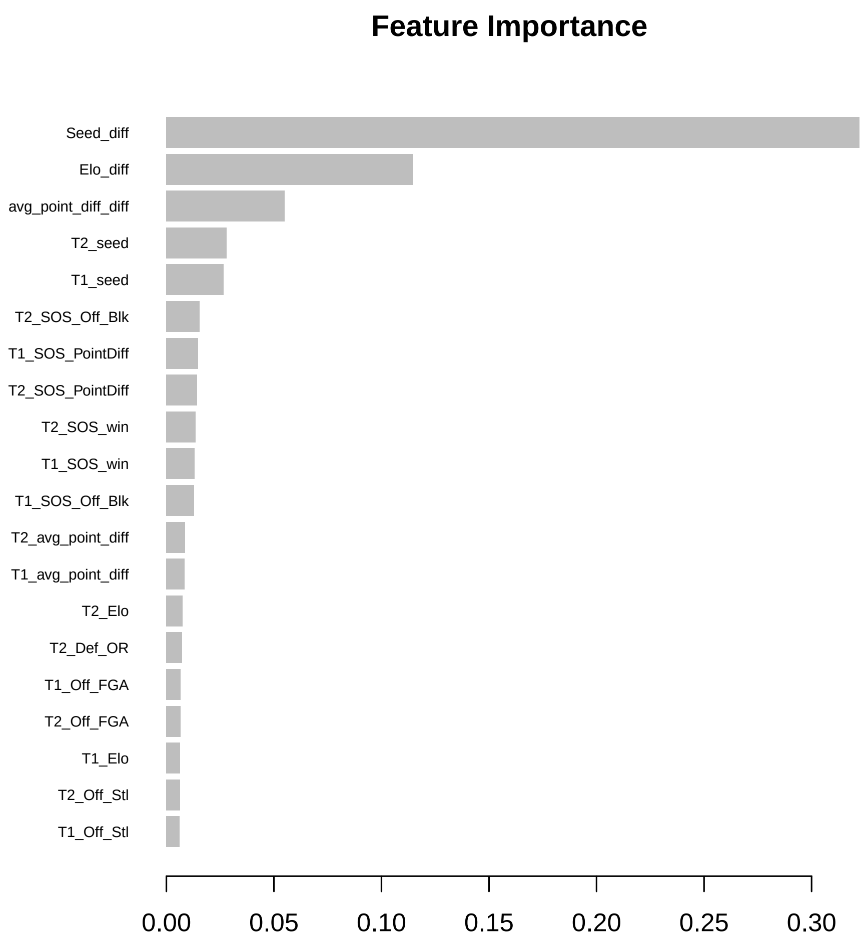

Figure 7. Feature Importance for the XGBoost Model.

Results

The performance of the XGBoost model on the 2025 NCAA tournament is evaluated along two dimensions: classification accuracy via the confusion matrix and probability calibration via the Brier Score. Figure 8 presents the confusion matrices for the men's, women's, and combined results.

The women's model achieves the strongest classification performance at 80.6% accuracy, followed by the combined model at 79.1% and the men's model at 77.6%. This ordering is consistent with the feature importance analysis, where broad measures of team quality such as seeding and Elo drive predictions, and the women's tournament historically exhibits stronger seed correlation with outcomes than the men's. All three accuracy figures comfortably exceed the 50% baseline expected from random prediction and improve upon the logistic regression accuracies of 77.6%, 73.9%, and 70.1% respectively, confirming that the expanded feature set and nonlinear modeling framework contribute meaningful predictive gains.

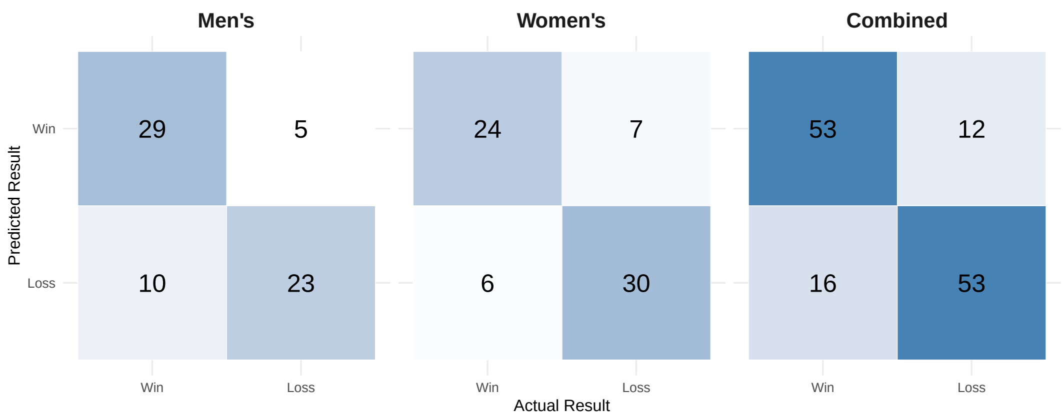

Figure 8. Confusion Matrices for the 2025 XGBoost Model

The combined 2025 Brier Score of 0.12761 reflects solid probability calibration but falls short of the competition-winning score of 0.10411. While the model correctly identifies the winner in the majority of matchups, the gap in Brier Score suggests that the predicted probabilities are not as sharply calibrated as those produced by the top Kaggle submissions, which typically incorporate additional data sources and more extensive tuning. Nevertheless, the XGBoost model represents a meaningful improvement over the logistic regression Brier Score of 0.18401, demonstrating that the richer feature set and ensemble framework produce better-calibrated win probabilities in addition to higher classification accuracy.

Point Spread Models

The logistic regression and XGBoost models developed in the preceding sections predict win probabilities directly from features derived entirely from the Kaggle competition dataset. While both models demonstrated meaningful predictive performance, they share a fundamental limitation: the outcome variable is modeled as a function of aggregated season statistics without any explicit representation of how many points each team is expected to score in a given matchup. The Linear Regression Monte Carlo model takes a different approach by decomposing the prediction problem into two stages. In the first stage, a linear regression model is trained to predict the number of points a team will score given its own offensive efficiency, its opponent's defensive efficiency, and the pace at which both teams play. In the second stage, the predicted scores and their associated uncertainty are used to simulate each game thousands of times, with the win probability derived as the proportion of simulations in which the team of interest outscores its opponent. This simulation-based framework incorporates external efficiency ratings from two sources: KenPom for the men's division and Bart Torvik for the women's division. Both sources provide opponent-adjusted offensive and defensive efficiency metrics that are not available in the raw Kaggle box score data, allowing the model to draw on a richer and more carefully constructed measure of team quality than the previous two models.

Linear Regression Monte Carlo

The Linear Regression Monte Carlo model is motivated by the observation that predicting a final score is a more tractable regression problem than directly estimating a win probability. Rather than asking the model to learn a complex nonlinear boundary between wins and losses, the linear regression stage asks a simpler question: given two teams' efficiency profiles, how many points should each team score? The residual standard deviation from this regression, denoted σ, captures the inherent unpredictability of college basketball scoring and serves as the engine of the simulation stage. By drawing repeated samples from the implied score distributions and counting the proportion of simulated games in which each team wins, the model converts a point estimate into a well-calibrated probability that naturally reflects the uncertainty in the underlying score prediction.

Variable Selection

The feature set for the Linear Regression Monte Carlo model is built around opponent-adjusted efficiency ratings sourced externally rather than computed from raw box score totals. For the men's division, KenPom provides adjusted offensive efficiency (`ORtg`), adjusted defensive efficiency (`DRtg`), and adjusted tempo (`AdjT`), all of which account for the strength of opposition faced and the pace at which games were played [10]. For the women's division, equivalent metrics are sourced from Bart Torvik, providing adjusted offensive efficiency (`AdjOE`), adjusted defensive efficiency (`AdjDE`), and adjusted tempo (`AdjT`) [3]. These opponent-adjusted metrics are preferred over the raw box score averages available in the Kaggle dataset because they isolate a team's true efficiency from the confounding influence of schedule strength, providing a cleaner signal for predicting how many points a team will score against a given opponent.

A key design decision in the feature construction is how the efficiency ratings are paired for each prediction. Rather than inputting each team's statistics in isolation, the model explicitly pits each team's offensive profile against the opponent's defensive profile. To predict Team A's score, the model uses Team A's adjusted offensive efficiency as `OffRating` and Team B's adjusted defensive efficiency as `DefRating`, along with both teams' tempo values. The same logic applies symmetrically to predict Team B's score. This pairing structure mirrors how scoring actually occurs in basketball — a team's expected output is not simply a function of how well it attacks, but how well it attacks against a specific opponent's defense. A team with an elite offense facing an elite defense should be expected to score fewer points than the same team facing a weak defense, and the feature pairing ensures the model captures this interaction directly rather than leaving it to be inferred from aggregate statistics alone.

Two additional differential features are included to capture relative team quality beyond the efficiency ratings alone. The Elo differential (`Diff_Elo`), computed as the difference between each team's end-of-season Elo rating, reflects the cumulative outcome of each team's season relative to its opponents. The Elo rating system used here follows the implementation described in [7], originally written in Python and reimplemented in R for this analysis.

For the men's division, an additional net rating differential (`Diff_NetRtg`) is also included, defined as the difference between each team's KenPom net rating, which is the margin between adjusted offensive and defensive efficiency. This feature is not included in the women's model because the Bart Torvik data source does not provide a precomputed net rating, and computing it manually from `AdjOE` and `AdjDE` was not performed during implementation. This represents a known limitation of the women's model relative to the men's, and future iterations should compute and include this feature to fully align the two frameworks. Season identifiers and team identifiers are excluded from the feature matrix as they carry no generalizable predictive signal.

Data Preparation

The data preparation pipeline for the Linear Regression Monte Carlo model differs from the previous two models in two important ways. First, the training data is drawn exclusively from the current regular season rather than historical tournament games. The men's model is trained on all 2025 regular season games and the women's model is trained on all 2025 regular season games, meaning the linear regression learns to predict points scored from efficiency ratings and pace as observed in that season's regular season play. This is appropriate because the external efficiency ratings from KenPom and Bart Torvik are also season-specific, reflecting the adjusted efficiencies accumulated over the 2025 regular season only. Second, the model requires an additional data integration step not present in the previous models: team name matching between the external ratings sources and the Kaggle team identifier system.

KenPom and Bart Torvik use their own team name conventions which do not always align with the spellings used in the Kaggle dataset. To resolve this, a two-stage matching pipeline is applied. In the first stage, team names from the external source are normalized to lowercase and matched exactly against the Kaggle team spellings lookup file. Teams that match exactly are assigned their Kaggle `TeamID` directly. In the second stage, teams that remain unmatched after exact matching are passed through a fuzzy matching procedure using the Jaro-Winkler string distance metric with a maximum distance threshold of 0.3 [30]. For each unmatched team the closest spelling in the Kaggle lookup file is selected, allowing minor variations in punctuation, abbreviation, and spacing to be resolved automatically. This two-stage approach ensures that the efficiency ratings can be reliably joined to the Kaggle game results without manual intervention. In practice all teams in both the men's and women's divisions were successfully matched, with no teams remaining unmatched after the two stage pipeline.

Each regular season game is duplicated symmetrically in the same manner as the XGBoost training data, generating one row from the scoring team's perspective and one from the opponent's perspective. This ensures the model learns to predict points from both orientations and does not develop a directional bias. The Elo ratings used as a differential feature are computed from the full historical regular season results available in the Kaggle dataset, spanning multiple decades of game results. At the start of the first available season every team is initialized at 1500. Ratings are then updated after every regular season game and carried over between seasons according to $R_{\text{new}} = 0.5 \times R_{\text{final}} + 750$, meaning each team's end-of-season rating partially persists into the following year rather than resetting. By the time the 2025 end-of-season ratings are extracted, they reflect the cumulative outcome of over two decades of regular season results, giving historically dominant programs a meaningfully higher baseline than programs with limited historical success. Following symmetric duplication, the training matrix is filtered using `complete.cases` to retain only rows where every feature is present. In practice no rows were removed, indicating that every team in the 2025 regular season had sufficient data across all feature sources to produce a complete profile. The final men's training matrix consisted of $10,326$ rows and 6 features, while the women's training matrix consisted of $10,828$ rows and 5 features, with the women's model excluding `Diff_NetRtg` as discussed in the Variable Selection section.

Model Implementation

The linear regression model is fit using ordinary least squares, which identifies the coefficient vector $\hat{\beta}$ that minimizes the sum of squared residuals between the predicted and observed scores across all training observations. Unlike the logistic regression described in the prior section, which models the log-odds of a binary outcome, the linear regression here models a continuous response, the number of points a team scores in a given game. The formal specification for the men's model is:

$$\begin{aligned}\text{Points} &= \beta_0 + \beta_1(\texttt{OffRating}) + \beta_2(\texttt{DefRating}) + \beta_3(\texttt{AdjT\_team}) \\ &\quad + \beta_4(\texttt{AdjT\_opp}) + \beta_5(\texttt{Diff\_NetRtg}) + \beta_6(\texttt{Diff\_Elo}) + \varepsilon\end{aligned} \tag{15}$$

and for the women's model:

$$\begin{aligned}\text{Points} &= \beta_0 + \beta_1(\texttt{OffRating}) + \beta_2(\texttt{DefRating}) + \beta_3(\texttt{AdjT\_team}) \\ &\quad + \beta_4(\texttt{AdjT\_opp}) + \beta_5(\texttt{Diff\_Elo}) + \varepsilon\end{aligned} \tag{16}$$

where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is the residual error term. The residual standard deviation σ is estimated from the training data and extracted as `summary(model)$sigma`. The values of σ for both divisions are reported and interpreted in the Model Analysis section, where they serve as a measure of the model's predictive accuracy. Here σ serves as the key input to the simulation stage, quantifying the uncertainty around each predicted score. The resulting coefficients reflect both the direction and magnitude of each predictor's contribution to the expected score. A positive coefficient on `OffRating` indicates that a team with a higher adjusted offensive efficiency is expected to score more points, which is consistent with basketball theory. Similarly, a positive coefficient on `DefRating` reflects the opponent's defensive efficiency, a higher opponent defensive rating means the opponent allows more points per possession, which naturally increases the scoring team's expected output. A positive coefficient on `AdjT_team` indicates that teams playing at a faster pace are expected to score more points in absolute terms, since more possessions create more scoring opportunities, while `AdjT_opp` captures the opponent's contribution to the pace of the game.

To convert the predicted scores into win probabilities, a Monte Carlo simulation is applied to each tournament matchup. For a given matchup between Team A and Team B, the linear regression produces two point estimates $\hat{\mu}_A$ and $\hat{\mu}_B$, representing the expected scores for each team given their respective efficiency profiles and the matchup context. Score realizations are then drawn independently for each team:

$$\text{Score}_A \sim \mathcal{N}(\hat{\mu}_A, \sigma^2) \tag{17}$$

$$\text{Score}_B \sim \mathcal{N}(\hat{\mu}_B, \sigma^2). \tag{18}$$

The win probability for Team A is then computed as:

$$\hat{P}(\text{Team A wins}) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[\text{Score}_{A,i} > \text{Score}_{B,i}] \tag{19}$$

This proportion $\hat{P}$(Team A wins) is submitted directly to Kaggle as the predicted win probability for the matchup, with the complementary probability $1 - \hat{P}$(Team A wins) implicitly assigned to Team B. The normality assumption for score distributions is motivated by the central limit theorem: a team's final score is the aggregate of many independent possession outcomes, and the distribution of these aggregates converges toward normality as the number of possessions increases. The men's model uses

$n = 100,000$ simulations per matchup, and the women's 2025 submission used $n = 50,000$ due to an implementation inconsistency discussed in the limitations section. At both sample sizes, the standard error of the proportion estimate is bounded by $\sqrt{p(1-p)/n} \leq 0.002$ at $n = 50,000$ in the worst case, confirming that the estimates have converged to stable values.

As with the logistic regression, separate models are trained for the men's and women's divisions rather than a single combined model. The justification mirrors that of the prior section. The relationship between efficiency ratings, tempo, and scoring may differ structurally between divisions due to differences in pace, physicality, and playing style. Training separate models allows each set of coefficients to reflect the unique dynamics of its respective division rather than forcing shared parameter estimates that could mask meaningful differences.

An important practical consideration in this implementation is the decision to train exclusively on the current regular season rather than pooling multiple years of data. A single NCAA regular season produces several thousand games per division, which is sufficient to estimate a low-dimensional model with six or seven predictors reliably. However, this also means the model is sensitive to season-specific variation in how efficiency ratings translate to scoring. A team whose KenPom ratings were inflated by a weak schedule may be overestimated by the model, and conversely, a team that peaked late in the season after an early slump may have efficiency ratings that understate their true tournament form. These limitations are inherent to any model that relies on season-aggregate statistics and are shared to varying degrees by all three modeling approaches in this analysis. For this reason, the model is deliberately kept parsimonious, relying on six predictors for the men's model and five for the women's rather than expanding the feature set further. This is consistent with the principle of Occam's Razor applied in the logistic regression section. A simpler model that generalizes reliably is preferred over a more complex one that risks overfitting to season-specific patterns in the training data [23].

Comparison of models

pros and cons of each model. going into detail why we pick the model we do for the tournament talking about the results of each model in the competition, as well as our submission.

Strategies

Here are my insights...

Results

CONCLUSIONS

i am free

Summary

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